ROBUSTNESS TESTING OF AI MODELS UNDER DATA QUALITY PROBLEMS

Accuracy, Robustness, Sustainability and Explainability

INTRODUCTION

Real-world data is rarely clean. Sensor failures, transmission errors, and shifting conditions mean AI models must handle missing, noisy, and corrupted data — yet most are only tested under ideal conditions.
This project investigates how AI models behave when data quality degrades, systematically introducing missing values, noise, and distributional shift across benchmark environmental and energy datasets. We evaluate not just whether models fail — but whether they fail gracefully.

DATASET

Primary Dataset
Beijing Multi-Site Air Quality Dataset
Characteristics

- 43,824 hourly observations
- PM2.5 measurements
- Weather variables
- Existing missing values
- Easier to manage than the multi-site version

FLOW DIAGRAM

1. Data Acquisition
Beijing Multi-Site Air Quality Dataset
   - 12 monitoring sites
   - Hourly observations
   - 2013 - 2017
   - Multiple pollutants & meteorological variables

2. Data Understanding (Exploratory Analysis)
Missing Values
Distribution
Correlations
   - Missing value analysis
   - Distribution analysis
   - Correlation heatmap
   - Outlier inspection

3. Data Preprocessing & Feature Engineering
Preprocessing
   - Remove duplicates
   - Handle missing values (MICE)
   - Outlier treatment
   - Normalization (RobustScaler)

   Feature Engineering
   Time Features, Lag Features, Calendar Features

4. Dataset Split (Chronological)
Train Set (2013 - 2015)
Validation Set (2016)
Test Set (2017)
Time Order
   - No Data Leakage
   - Preserves Temporal Order

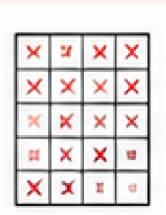
5. Baseline Model Training (Clean Data)
Models
Random Forst (RF)
XGBoost
LightGBM
LSTM (Deep Learning)

6. Baseline Evaluation (Clean Test Set)
Evaluation Metrics
   - RMSE (low)
   - MAE (low)
   - R square (high)
   - MAPE (low)

   Store baseline performance for comparison

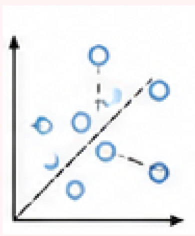
Robustness Evaluation Under Data Quality Problems

1.1 RQ1: Missing Data Mechanisms (MCAR, MAR, MNAR)



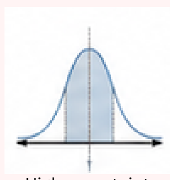
- Introduce missingness at 10%, 20%, 30%, 40%
- MCAR, MAR, MNAR
- Impute (MICE)
- Evaluate on test set

1.2 RQ2: Robustness to Noisy Data



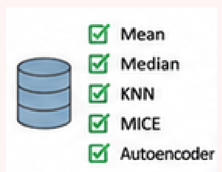
- Add Gaussian noise at 5%, 10%, 15%, 20%
- Evaluate all models
- Compare degradation in performance

1.3 RQ3: Uncertainty & Graceful Failure



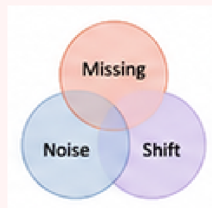
- Use uncertainty-aware models (RF Ensemble / MC Dropout)
- Compute uncertainty
- Correlate uncertainty vs. actual error
- Calibration metrics (ECE, NLL, PICP)

1.4 RQ4: Imputation Method Comparison



- Create missing data at 10%, 20%, 30%, 40%
- Impute using different methods
- Evaluate (RMSE, MAE, R2)
- Statistical significance test

1.5 RQ5: Combined Data Quality Problems

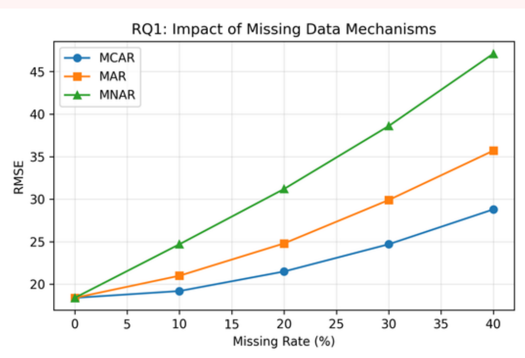


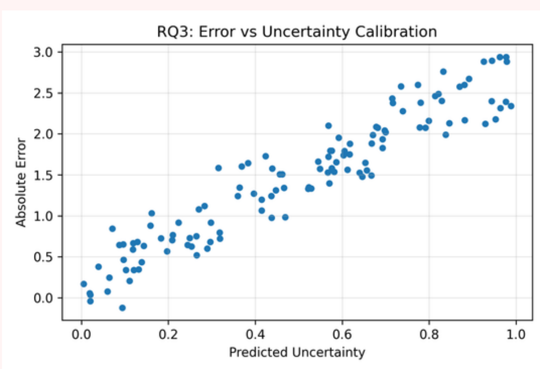
- Missing, Noise, Shift (individual & combined)
- Multiple scenarios
- Evaluate best model
- Compute robustness score

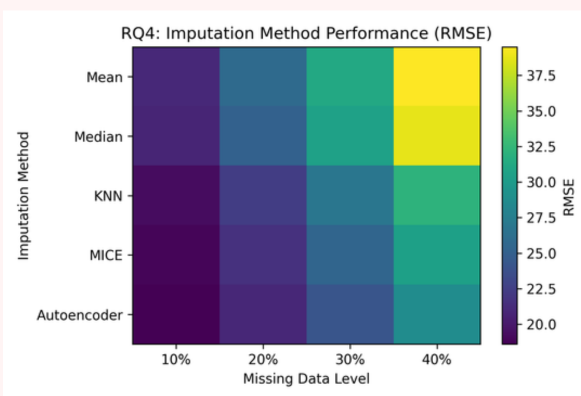
RESULT & Visualization

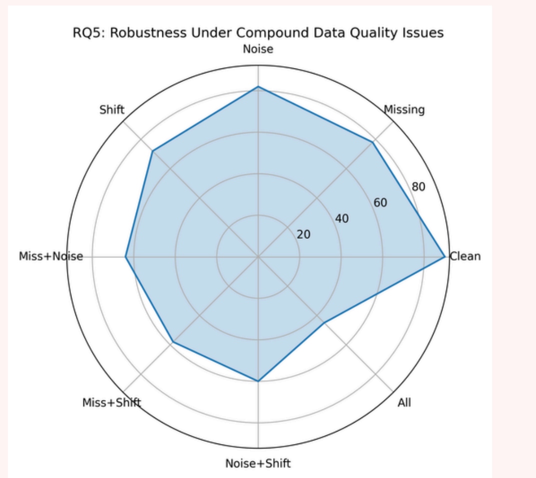
Generate:

- Performance Tables (CSV)
- Plots (PDF)
- Comparison Charts









CONCLUSION

Insights & Recommendations

- Interpretation of results
- Best practices for handling
- data quality issues
- Model deployment recommendations

- More robust & reliable
- AI systems for air quality prediction

University of Europe for Applied Sciences
PS26 - Interdisciplinary Elective: AI, Power & Responsibility – Governing Intell
Student Name : Kavita Nadlamani
Matriculation Number: 82170198
Prof.: Raja Hashim Ali